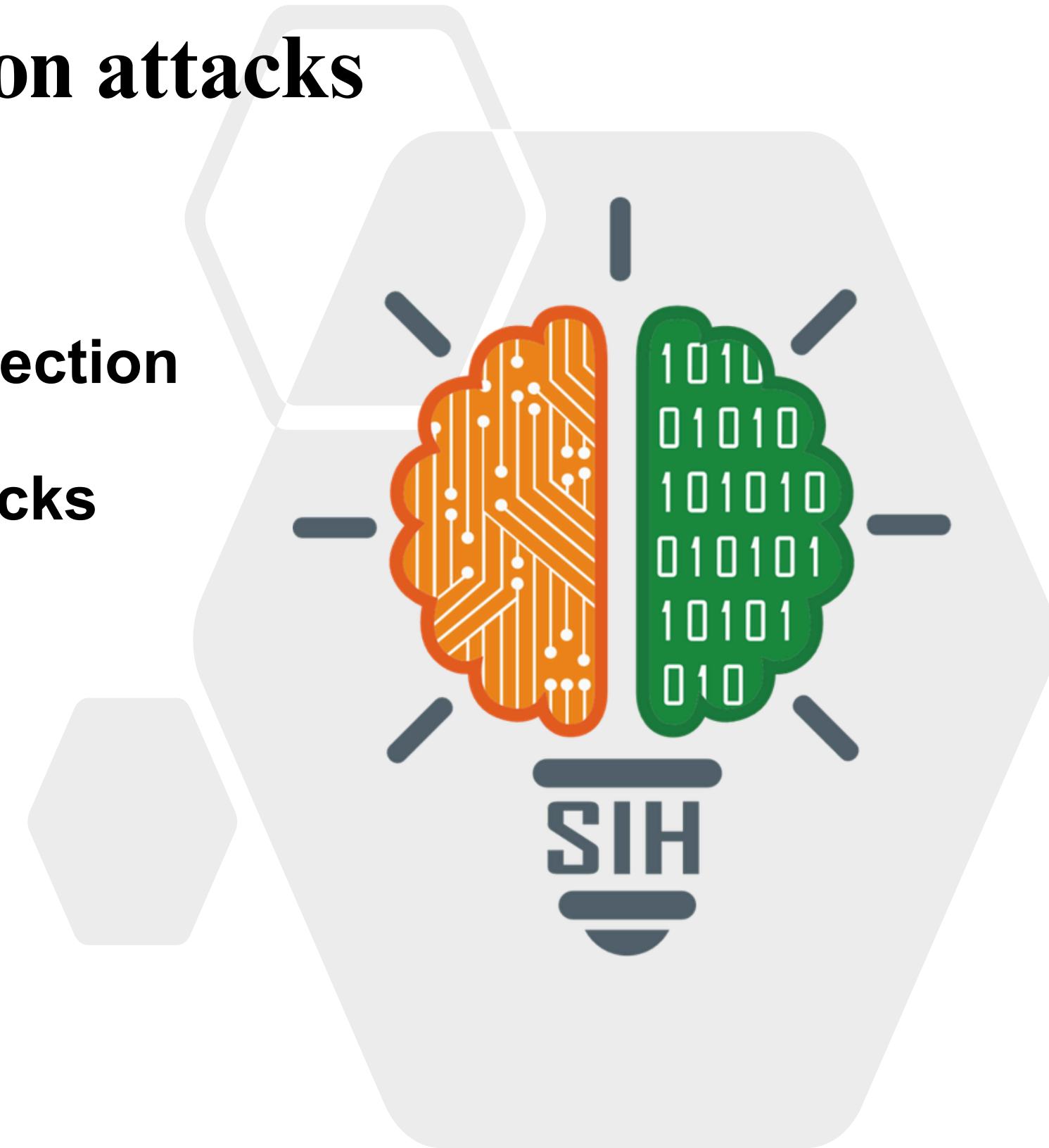


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AI-Powered Real-Time Detection and Prevention of Voice cloning impersonation attacks

- Problem Statement ID – SIH26104
- Problem Statement Title- AI-Powered Real-Time Detection and Prevention of Voice cloning impersonation attacks
- Theme- Blockchain & Cybersecurity
- PS Category- Software
- Team ID-
- Team Name- VoxGuard





VoxGuard.AI

Detect. Verify. Prevent.
Protect Every Voice.



VoxGuard.AI

AI-Powered Real-Time Detection and Prevention of
Voice Cloning Impersonation Attacks

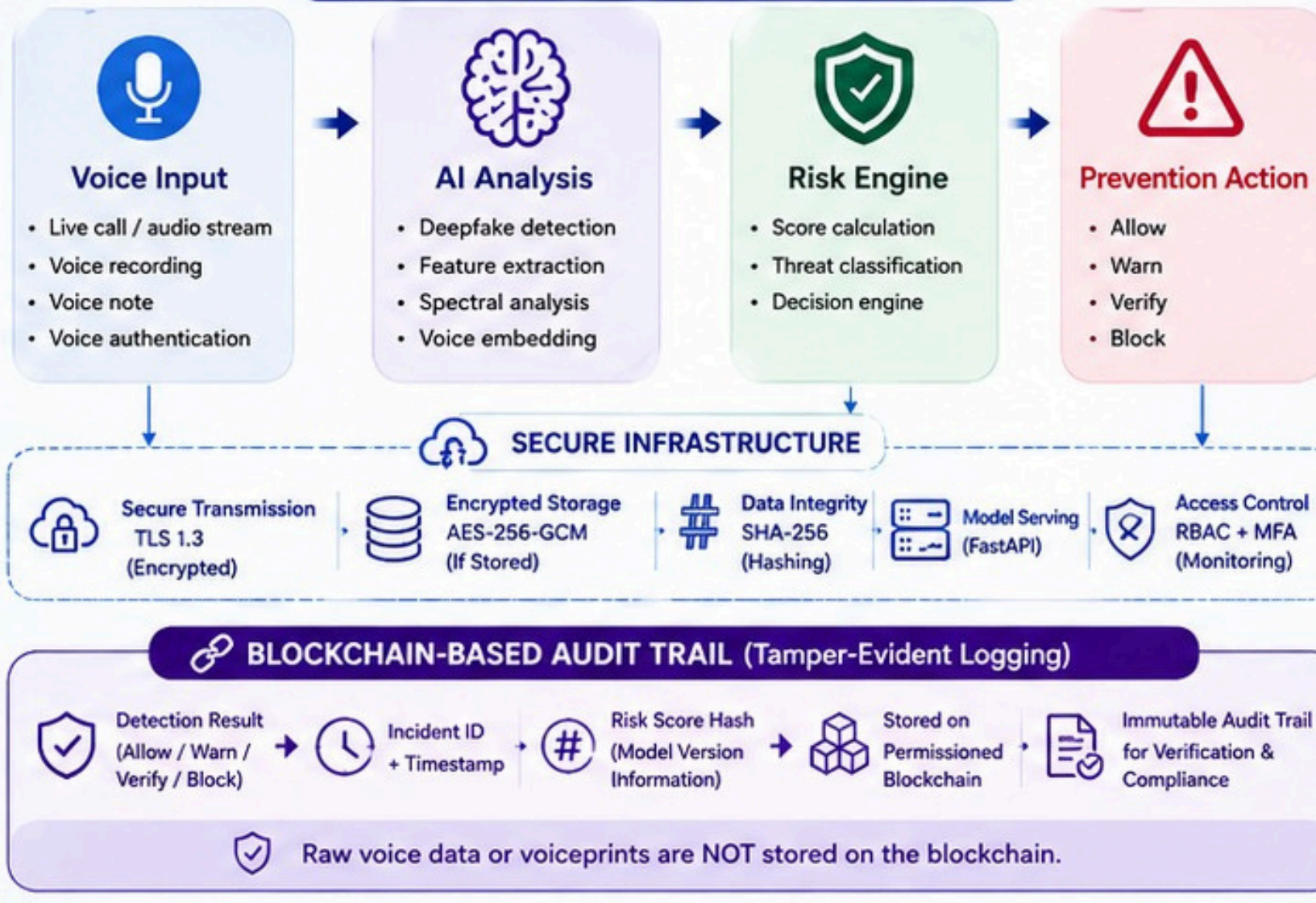


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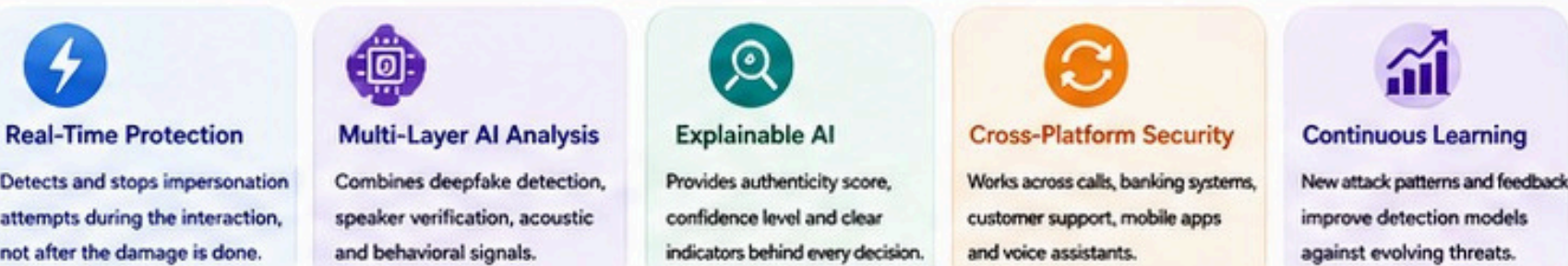
6 - PILLAR PROTECTION FRAMEWORK



VOICE AUTHENTICITY ENGINE



INNOVATION AND UNIQUENESS OF THE SOLUTION



KEY SOLUTION MODULES





VoxGuard.AI

Detect. Verify. Prevent.
Protect Every Voice.



TECHNICAL APPROACH

AI-Powered Real-Time Detection and Prevention of
Voice Cloning Impersonation Attacks



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1. INPUT PROCESSING LAYER WORKFLOW



AUDIO CAPTURE

- Captures live audio / voice stream in real time.



AUDIO PREPROCESSING

- Noise reduction
- Voice Activity Detection (VAD)
- Silence trimming
- Normalization & resampling



FEATURE EXTRACTION

- Mel-Spectrogram
- MFCC
- Spectral & temporal features
- Pitch, energy, formants



QUALITY & INTEGRITY CHECK

- Detect compression artifacts
- Detect replay attacks
- Signal quality assessment



OUTPUT

Feature-enriched audio features ready for ML analysis.

2. AI/ML DECISION ENGINE

A. DEEPPAKE DETECTION MODEL

CNN / EfficientNet / ResNet based models detect synthetic vs. real based on deep learned signals.



B. SPEAKER VERIFICATION

ECAPA-TDNN / Speaker Embedding verify voice identity against enrolled voiceprints.



C. ANTI-SPOOFING ENGINE

Detects replay attacks, TTS, voice conversion or cloned voices using liveness techniques.



D. ACOUSTIC & BEHAVIOR ANALYSIS

Analyzes acoustic patterns, speaking behavior, linguistic inconsistencies & prosody shifts.



E. ENSEMBLE RISK ENGINE

Combines outputs of all modules using weighted fusion to generate final authenticity score and risk level.

3. RISK ASSESSMENT & RESPONSE

LOW RISK



Voice appears genuine
→ Allow interaction

MEDIUM RISK



Minor anomalies detected
→ Trigger warning

HIGH RISK



Strong synthetic indicators
→ Require verification

CRITICAL RISK



High probability of impersonation
→ Block action & alert user/admin

4. FEEDBACK LOOP & CONTINUOUS LEARNING



Suspicious Cases & User Feedback



Store & Tagging of Samples



Model Retraining & Fine-Tuning



Improved Models & Better Detection

5. BACKEND & INFRASTRUCTURE

BACKEND SERVICES



FastAPI (Python)
AI/ML & model serving



WebSocket
Real-time audio streaming



REST APIs
Application integration



STORAGE

Encrypted storage for audio samples (optional), logs, model & reports.



DATABASE

PostgreSQL / Supabase

- User profiles
- Voice embeddings
- Detection history
- Risk logs
- Alerts & reports



BLOCKCHAIN-BASED AUDIT LOGGING

- Blockchain-based tamper-evident audit logging
- Stores: Detection result + Incident ID + Timestamp + Risk score hash
- Raw voice data is NOT stored on blockchain



6. FRONTEND



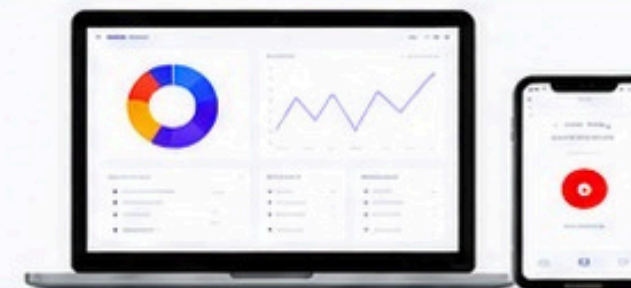
Web Dashboard
React / Next.js



Mobile Application
Flutter / React Native



Admin Panel
Monitoring & analytics



7. INTEGRATION & AUTOMATION



VoIP / Call Integration (Twilio, Asterisk)
Works with banks, call centers, apps, websites & enterprise platforms.



SIEM / Security Tools Integration
Auto-incident ticketing & alerts



Compliance reporting (GDPR, ISO, etc.)



Audit trail via blockchain ledger



End-to-End Security & Encryption



Privacy by Design
(No raw voice stored)



Blockchain Audit Trail for all high-risk cases



Real-Time Processing with Low Latency



Adaptive & Self-Learning AI Models



1. FEASIBILITY



TECHNICAL FEASIBILITY

- Uses proven AI/ML techniques
- Leverages open-source models & datasets
- Real-time processing achievable with optimized pipelines
- Scalable with cloud & edge deployment



OPERATIONAL FEASIBILITY

- Easy integration via APIs
- Minimal user training required
- Works across devices & platforms
- Automated mechanism & alerts reduce manual effort



DATA FEASIBILITY

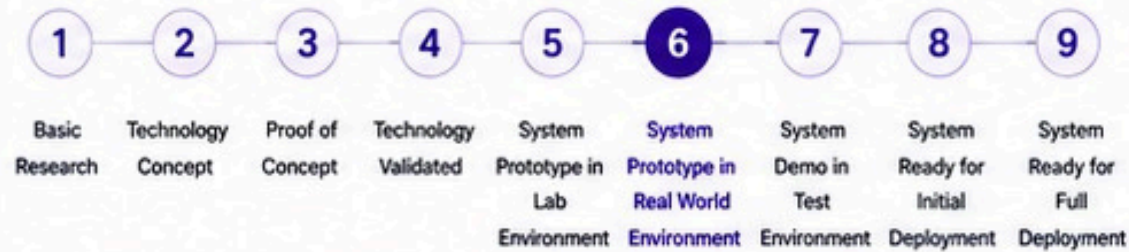
- Availability of public voice datasets (ASVspoof, VoxCeleb, LibriSpeech)
- Can be fine-tuned with organization specific data
- Data augmentation ensures robustness



RESOURCE FEASIBILITY

- Cloud + hybrid deployment
- Computationally cost-effective cloud resources
- Modular system makes it cost-effective
- Scales with demand & can be launched with lean team

TECHNOLOGY READINESS LEVEL (TRL)



KEY ENABLERS



2. VIABILITY



PROBLEM-SOLUTION FIT

- Addresses a critical & growing threat – voice cloning impersonation
- Solves real-world problems for individuals, enterprises and government
- High demand for fraud prevention and identity assurance



MARKET POTENTIAL

- Global voice biometrics market expected to reach \$3.6B by 2030*
- Rising digital frauds and deepfake attacks
- Multiple sectors can benefit – Banking, Telecom, E-commerce, Govt., Healthcare



SUSTAINABILITY

- Scalable SaaS cloud model
- Continuous learning adapts to new attack patterns
- Efficient resource utilization & cost optimization
- Environment-friendly with digital practices



SOCIAL IMPACT

- Protects users from financial fraud & identity theft
- Builds digital trust and security
- Encourages innovation in secure voice-based interactions
- Contributes to a safer digital world



BUSINESS & REVENUE MODEL

- ✓ Freemium model for individuals & small businesses
- ✓ Tiered pricing for enterprises
- ✓ AI model integration for platforms & apps
- ✓ Custom deployments for government & large organizations



COST STRUCTURE (Efficient & Scalable)

- Cloud Infrastructure
- Model Training & Inference
- Storage & Data Processing
- Maintenance & Support
- Continuous R&D

WHY IT IS FEASIBLE & VIABLE



Built on proven technologies and open-source tools



Strong market need and scalability potential



Experienced & motivated student team



Achievable within hackathon timeline with real impact



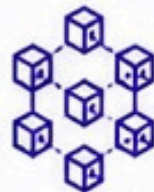
High potential for real-world adoption and growth

ADDITIONAL SECURITY & TRUST ENABLERS



CYBERSECURITY BY DESIGN

End-to-end encryption, secure APIs, RBAC & MFA for access control



BLOCKCHAIN-BASED AUDIT TRAIL

Tamper-evident, immutable logging of critical events & decisions



PRIVACY BY DESIGN

Raw voice data is NOT stored on blockchain or in plain form



COMPLIANCE READY

Built aligned with GDPR, ISO 27001, and data protection best practices



SCALABLE & FUTURE-READY

Modular architecture that adapts to new threats & technologies



VoxGuard.AI

Detect. Verify. Prevent.
Protect Every Voice.



IMPACT & BENEFITS



Stronger Voices. Safer Conversations. A More Secure Tomorrow.



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1. KEY IMPACT AREAS



Financial Protection

Prevents fraud and monetary loss by stopping impersonation and scams.



Identity & Privacy

Protects personal identity and keeps voice data secure.



Digital Trust

Builds confidence in voice-based communication and digital platforms.



Social Safety

Protects vulnerable groups (elderly, children, etc.) from voice impersonation abuse.



National Security

Helps prevent misuse of cloned voices in critical communications.

2. HOW VOXGUARD.AI CREATES IMPACT



1 Detects Voice Cloning in Real-Time

AI models analyze voice patterns instantly to detect synthetic or cloned audio.



2 Blocks Impersonation Attempts

Suspicious calls/transactions are blocked before any damage occurs.



3 Assesses Risk Accurately

Risk engine evaluates confidence scores and provides clear risk level.



4 Alerts & Notifies Instantly

Users/admins receive instant alerts to take preventive action.



5 Enables Safer Communication

Ensures genuine, trusted and secure voice interactions for everyone.

3. WHO BENEFITS?



Individuals

Personal safety, identity protection and peace of mind.



Banks & Financial Institutions

Reduces fraud, chargebacks and financial losses.



Enterprises & Organizations

Protects internal communication and prevents data breaches.



Call Centers & Support Teams

Filters fake calls and improves customer trust.



Government & Public Services

Secures citizen interactions and critical voice communications.

4. OVERALL BENEFITS & OUTCOMES



Real-Time Protection

Detects and blocks attacks as they happen.



Reduced Fraud & Losses

Minimizes financial and identity theft.



Stronger Digital Trust

Builds confidence in platforms and voice communication.



Scalable & Reliable

Works across industries, platforms and use cases.

5. IMPACT AT A GLANCE



Up to
95%+

Detection Accuracy



70%+

Reduction in Fraud Attempts



Real-Time
Detection in
< 2 Seconds



Safer for
Millions
of Users



Applicable Across
Multiple Sectors
and Use Cases

6. LONG-TERM SOCIETAL IMPACT



Creates a safer
digital ecosystem



Reduces cybercrime
and digital fraud



Promotes ethical
and trusted AI



Contributes to a
secure digital future
for all



VoxGuard.AI empowers a future where every voice is real, every conversation is trusted, and every identity is protected.

Stronger Voices. Safer Conversations. A More Secure Tomorrow.





RESEARCH & REFERENCES



RESEARCH AREA		REFERENCES / LINKS	
	Deepfake Voice Detection Research on methods and challenges in detecting synthetic and manipulated voices.		https://ieeexplore.ieee.org/document/10811921
	Speaker Verification & Anti-Spoofing Studies on speaker verification and anti-spoofing techniques and evaluation methods.		https://ieeexplore.ieee.org/document/10918392
	Voice Cloning Attack Detection Deep learning based approaches for detecting voice cloning and impersonation attacks.		https://ieeexplore.ieee.org/document/10890861
	Explainable / Real-Time Anti-Spoofing Explainable AI and real-time frameworks for voice anti-spoofing and biometric security.		https://ieeexplore.ieee.org/document/10569111
	ASVspooof & Countermeasure Research Benchmark datasets and countermeasures for automatic speaker verification spoofing attacks.		https://ieeexplore.ieee.org/document/10678901